

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 8

Vector Spaces and the Meaning of “Canvas”

Chapter 8 — Phase III: The Grammar of Structure

Self-Study Curriculum

The Reframe

- You already run linear algebra fluently: covariance, PCA, weight matrices.
- This chapter does not teach it; it makes the choices inside it visible.
- Chapter 7 left a thread: representations act on a vector space. What is that?
- The payoff is your own intuition, made precise: linearity lives in the canvas.

What a Vector Space Is

- A set of objects you can *add* and *scale*, obeying a short list of axioms.
- The axioms never say “arrow” or “column of numbers.”
- So the only membership test is: closed under adding and scaling.
- It is a type of *structure*, not a type of object.

Vectors That Are Not Columns

- Functions $\mathbb{R} \rightarrow \mathbb{R}$: add their values, scale them. Zero is the zero function.
- Solutions of a linear differential equation (this is why superposition works).
- Quantum states: adding and scaling is the physics, not a metaphor.
- Polynomials of degree at most 3: a 4-dimensional space.

Basis and Dimension

- A basis is a minimal kit of building blocks; dimension is its size.
- Same dimension for every basis: a fact about the space, not the description.
- **A choice of basis is a choice of description, not a fact about the space.**
- Coordinates change with the basis; the vector does not move.

Worked: One Polynomial, Two Frames

- $2x^3 - x + 5$ in the basis $\{1, x, x^2, x^3\}$: coordinates (5, -1, 0, 2).
- In the basis $\{1, (x-1), (x-1)^2, (x-1)^3\}$: different coordinates (a Taylor report).
- The numbers changed completely; the polynomial is identical.
- Choosing features is choosing a basis. Re-encoding moves coordinates, not the data.

Linear Maps

- Preserve addition and scaling: $f(u+v) = f(u)+f(v)$, $f(cv) = cf(v)$.
- The third instance of the Chapter 7 theme (operation, addition/scaling, nearness).
- A matrix is the *coordinate report* of a map once bases are chosen.
- Change the basis and the matrix changes; the map does not.

Eigenvalues Are the Invariants

- The matrix is an artifact of your basis. What is real about the map?
- Eigenvalues: scaling factors along special directions, unchanged by basis.
- They belong to the map, not the coordinates (like the determinant, Chapter 5).
- Covariance eigenvalues = variance along principal directions, coordinate-free.

Linearity Lives in the Canvas

- Choose a representation (encode the state as a vector): that is the canvas.
- On that canvas the evolution is either linear (matrix multiplication) or not.
- A different canvas can flip the same evolution linear or nonlinear.
- Linearity is a fact about system-and-canvas, not about the system alone.

Worked: Population Growth

- Count $N(t)$: the growth curve $N(t) = N_0 e^{rt}$ *bends* when plotted against time.
- Log canvas: $\log N(t) = \log N_0 + rt$ draws as a straight line.
- Same bacteria, same system; only the canvas changed the *shape of the plot*.
- Two senses of “linear”: as a rule, $dN/dt = rN$ already superposes on the N canvas; the log canvas straightens the *graph*, a different (everyday) sense.

Linearization

- No exact linear canvas? Use one that is linear in a small region.
- Linearizing near equilibrium replaces the map by its best linear stand-in.
- Assumption: deviations are small enough that superposition holds well.
- A flat canvas touching a curved system at one point (doorway to Chapter 10).

Inner Products Add Geometry

- Adding, scaling, and bases carry no length or angle.
- An inner product supplies length, angle, and orthogonality.
- Not every space has a natural one; when there is a choice, it matters.
- Different inner products make different vectors count as perpendicular.

PCA as a Change of Canvas

- Old canvas: 100 measured features, one axis each.
- New canvas: eigenvectors of the covariance, ordered by variance.
- Keep the top 3: dimension $100 \rightarrow 3$.
- Lost: 5% of variance and exact invertibility. Gained: compression.

Eigenvalues = variance per direction = the system's intrinsic dimensionality.

Which Analyst Is Right?

- Raw features vs PCA features, same algorithm, different results: neither is simply wrong.
- They assume different bases and inner products for the phenomenon's geometry.
- Rotate before clustering and clusters move: the algorithm is not rotation-invariant.
- The “right” inner product matches the system's own notion of similarity.

Takeaway and Bridge

- A vector space is a structure; a canvas is a basis plus an inner product.
- Linearity, coordinates, and matrices are canvas-dependent; eigenvalues are not.
- Buying geometry cost an inner product: distances and angles, and they are fragile.
- Chapter 9: throw all measurement away, keep only nearness. That is topology.